Math Thermo
class # 06
02/05/2026

Legendre Transforms and Helmholtz Free Energy

Defn (6.1): Suppose that $f:[a,b]\to\mathbb{R}$ is continuous. For every $p\in[c,d]$, define

(6.1) $$f^*(p) := \sup_{x\in[a,b]}\{xp - f(x)\}.$$

The function $f^*:[c,d]\to\mathbb{R}$ is called the Legendre transform of f.

Theorem (6.2): Suppose that $f\in C^2([0,\infty);\mathbb{R})$ with

$$f''(x)>0, \quad \forall x\in[0,\infty).$$

Then, for all $p\in \text{Range}(f')$,

(6.2) $$f^*(p) = x_p\cdot p - f(x_p),$$

where $x_p\in[0,\infty)$ is the unique solution to

$$f'(x_p)=p.$$

Proof: Let $p\in\text{Range}(f')$. $f'[0,\infty)\xrightarrow[\text{onto}]{1-1}\text{Range}(f')$.
For every $p\in\text{Range}(f')$, there is a unique $x_p\in[0,\infty)$ such that,

$$f'(x_p)=p,$$

Since f' is a strictly increasing function.

Fix $p \in \text{Range}(f')$. By Taylor's Theorem,

$$f(x) = f(x_p) + p(x - x_p) + \frac{1}{2} f''(\xi_p)(x - x_p)^2$$

for some ξ_p between x and x_p.

only x dependence

$$\sup_{0 \le x < \infty} \{x \cdot p - f(x)\}$$

$$= \sup_{0 \le x < \infty} \{x_p \cdot p - f(x_p) - \frac{1}{2} f''(\xi_p)(x - x_p)^2\}$$

$$= x_p \cdot p - f(x_p). \quad ///$$

Theorem (6.3): Suppose that $f \in C^2([0,\infty); \mathbb{R})$ with

$$f''(x) > 0, \quad \forall x \in [0,\infty).$$

(6.3) Then, $$[f^*]'(p) = [f']^{-1}(p), \quad [f^*]''(p) > 0,$$

for all $p \in \text{Range}(f')$.

Proof: Recall

$$f' : [0,\infty) \xrightarrow[\text{onto}]{1-1} \text{Range}(f')$$

Thus

$$f'([f']^{-1}(p)) = p, \quad \forall p \in \text{Range}(f').$$

Also, observe that $\frac{d}{dp}\left[[f']^{-1}\right] \in C(\text{Range}(f'); \mathbb{R})$.

Now,
$$\frac{d}{dp}\left[f^*(p)\right] = \frac{d}{dp}\left\{x_p \cdot p - f(x_p)\right\}$$
$$= \frac{d}{dp}\left\{[f']^{-1}(p)\cdot p - f\left([f']^{-1}(p)\right)\right\}$$
$$= p \cdot \frac{d}{dp}\left[[f']^{-1}\right](p)$$
$$+ [f']^{-1}(p)$$
$$- \underbrace{\frac{df}{dx}\left([f']^{-1}(p)\right)}_{=p} \cdot \frac{d}{dp}\left[[f']^{-1}\right](p)$$
$$= [f']^{-1}(p).$$

Now that this is established, we have
$$\frac{d^2}{dp^2}\left[f^*(p)\right] = \frac{d}{dp}\left[[f']^{-1}\right](p)$$
for all $p \in \text{Range}(f')$. Since $f'' \in C([0,\infty))$ and
$$f''(x) > 0, \quad \forall\, x \in [0,\infty),$$
$$\frac{d}{dp}\left[[f']^{-1}\right](p) > 0, \quad \forall\, p \in \text{Range}(f'). \quad ///$$

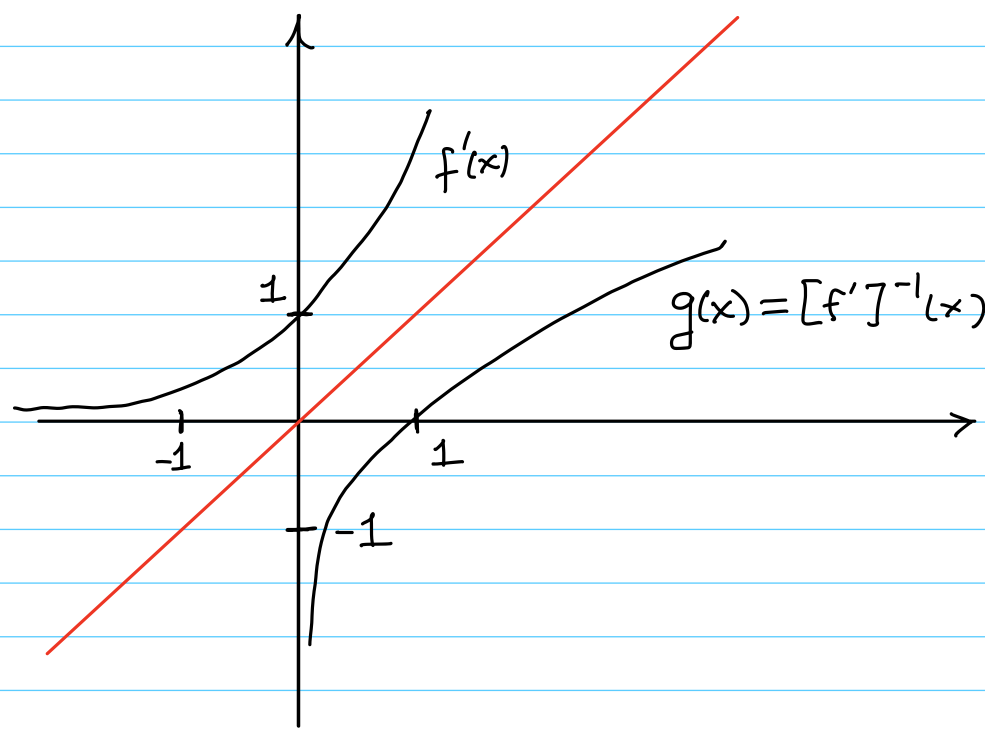
Example (6.4): Suppose that
$$f(x) = \exp(x), \quad x \in (-\infty, \infty).$$
We could restrict the domain to $[0,\infty)$, if desired.

Then,
$$f'(x) = \exp(x), \qquad f''(x) = \exp(x).$$

Clearly, $f'' > 0$ and f is convex. Let us examine the derivative function, f', because this is important

$$\mathrm{Dom}(f') = (-\infty, \infty)$$

$$\mathrm{Range}(f') = (0, \infty)$$



Clearly $f' : (-\infty, -\infty) \xrightarrow[\text{onto}]{1-1} (0, \infty)$ is an increasing function.

$$[f']^{-1} : (0, \infty) \xrightarrow[\text{onto}]{1-1} (-\infty, \infty)$$

is
$$[f']^{-1}(p) = g(p) = \log(p).$$

Observe that

$$f'(g(p)) = \exp(\log(p))$$
$$= p.$$

For all $p \in \text{Range}(f')$,

$$f^*(p) = x_p \cdot p - f(x_p),$$

where $x_p \in [0, \infty)$ is the unique solution to

$$f'(x_p) = p.$$

Thus,

$$f^*(p) = g(p)\,p - f(g(p))$$
$$= \log(p)\,p - \exp(\log(p)).$$
$$= p\log(p) - p,$$

for all $p \in \text{range}(f') = (0, \infty)$. This function

$$\boxed{p \mapsto p\log(p) - p}$$

is very important in Thermodynamics and is called the <u>ideal gas function</u>. More on this later.

We leave it to the reader to confirm that

$$[f^*]^*(x) = f(x) = \exp(x), \quad x \in \mathbb{R}.$$

We will see that this is true in general.

<u>Theorem</u> (6.5): Suppose that $f \in C^2([0,\infty);\mathbb{R})$ with

$$f''(x) > 0, \qquad \forall x \in [0,\infty).$$

Then $f^* \in C^2(\text{Range}(f');\mathbb{R})$ with

$$[f^*]''(p) > 0, \qquad \forall p \in \text{Range}(f').$$

Furthermore

(6.4) $$[f^*]^*(x) = f(x), \qquad \forall x \in [0,\infty).$$

In other words, the Legendre transform is involutory and information preserving.

<u>Proof</u>: It follows that, for all $x \in [0,\infty)$,

$$[f^*]^*(x) = x\,p_x - f^*(p_x),$$

where $p_x \in \text{Range}(f')$ is the unique solution to

$$f^{*\prime}(p_x) = x.$$

Recall

$$f^*(p_x) = p_x \cdot x_{p_x} - f(x_{p_x}),$$

where $x_{p_x} \in [0,\infty)$ is the unique solution to

$$f'(x_{p_x}) = p_x.$$

Of course, by uniqueness,

$$x_{p_x} = x.$$

So

$$[f^*]^*(x) = x\,p_x - f^*(p_x)$$

$$= x \cdot p_x - \{p_x \cdot x_{p_x} - f(x_{p_x})\}$$

$$= f(x). \quad ///$$

Why is the Legendre transform useful in Thermodynamics?

This transform allows us to introduce new thermodynamics coordinates / variables.

Recall

$$\tilde{U} = \tilde{U}(S, V, N)$$

or

$$d\tilde{U} = T\,dS - P\,dV + \mu\,dN$$

The latter means, in short hand,

$$T = \frac{\partial \tilde{U}}{\partial S}, \quad -P = \frac{\partial \tilde{U}}{\partial V}, \quad \mu = \frac{\partial \tilde{U}}{\partial N}.$$

$T, -P, \mu$ are called <u>conjugate variables</u>.

Extensive Quantity	Conjugate Variable	Variable
$\tilde{U}$	T	S
	$-P$	V
	μ	N
$\tilde{S}$	$1/T$	U
	P/T	V
	$-\mu/T$	N

Definition (6.6): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary ($r=1$). Define

(6.5) $$\tilde{F}(T,V,N) = \tilde{U}(S_T,V,N) - TS_T$$

where $S_T = S_T(V,N)$ is the unique solution of the equation

(6.6) $$T_U(S_T(V,N),V,N) = T$$

for fixed values of N and V.

$\tilde{F}$ is called the Helmholtz free energy.

Formally, we have

$$d\tilde{F} = d\tilde{U} - SdT - TdS$$

$$= TdS - PdV + \mu dN - SdT - TdS$$

$$= -SdT - PdV + \mu dN.$$

Extensive Quantity	Conjugate Variable	Variable
$\tilde{U}$	T	S
	$-P$	V
	μ	N
$\tilde{F}$	$-S$	T
	$-P$	V
	μ	N

Example (6.7): Suppose that

$$\tilde{S}(U,V,N) = \left(\frac{NVU R^2}{v_0 \theta}\right)^{1/3}.$$

It follows that

$$\tilde{U}(S,V,N) = \frac{S^3 v_0 \theta}{NVR^2}.$$

The temperature function is

$$T_U(S,V,N) = 3\frac{S^2 v_0 \theta}{NVR^2}.$$

Or

$$\frac{1}{T_S(U,V,N)} = \frac{1}{3}\left(\frac{NVU R^2}{v_0 \theta}\right)^{-2/3} \frac{NVR^2}{v_0 \theta}$$

$$= \frac{1}{3}\left(\frac{NVR^2}{v_0 \theta}\right)^{1/3} U^{-2/3}.$$

Thus

$$T_S(U,V,N) = 3\left(\frac{v_0 \theta}{NVR^2}\right)^{1/3} U^{2/3}.$$

Recall that

$$T_U(S,V,\vec{N}) = T_S(\tilde{U}(S,V,N),V,\vec{N})$$

and

$$T_S(U,V,\vec{N}) = T_U(\tilde{S}(U,V,\vec{N}),V,\vec{N}).$$

By definition

$$\tilde{F}(T,V,N) = \tilde{U}(S_T,V,N) - TS_T$$

where

$$T_U(S_T(V,N),V,N) = T.$$

For our example,

$$T_U(S,V,N) = 3\frac{S^2 v_0 \theta}{NVR^2}.$$

Hence,

$$S_T(V,N) = \sqrt{\frac{TNVR^2}{3v_0\theta}}.$$

Thus,

$$\tilde{F}(T,V,N) = \tilde{U}\left(\sqrt{\frac{TNVR^2}{3v_0\theta}}, V, N\right) - T\sqrt{\frac{TNVR^2}{3v_0\theta}}$$

But

$$\tilde{U}(S_T,V,N) = \frac{S_T^3 v_0 \theta}{NVR^2} = \left(\frac{TNVR^2}{3v_0\theta}\right)^{3/2}\frac{v_0\,\theta}{NVR^2}$$

So that

$$\tilde{F}(T,N,V) = \left(\frac{TNVR^2}{3v_0\theta}\right)^{3/2}\frac{v_0\,\theta}{NVR^2} - T\sqrt{\frac{TNVR^2}{3v_0\theta}}$$

$$= \frac{1}{3}\frac{T^{3/2}}{\sqrt{3}}\sqrt{\frac{NVR^2}{v_0\theta}} - \frac{T^{3/2}}{\sqrt{3}}\sqrt{\frac{NVR^2}{v_0\theta}}$$

Finally,

$$\boxed{\tilde{F}(T,N,V) = \frac{-2T^{3/2}}{3}\sqrt{\frac{NVR^2}{3v_0\theta}}.}$$

Now,

$$\frac{\partial\tilde{F}}{\partial T} = -\frac{2}{3}\cdot\frac{3}{2}\cdot\sqrt{T}\cdot\sqrt{\frac{NVR^2}{3v_0\theta}}.$$

But

$$T_U(S,V,N) = \frac{3S^2v_0\theta}{NVR^2}$$

Thus,

$$\sqrt{T} = \sqrt{\frac{3v_0\theta}{NVR^2}}\ S_T$$

and

$$\boxed{\frac{\partial\tilde{F}}{\partial T} = -S_T.}$$

Theorem (6.8): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary $(r=1)$. The following provides an equivalent means for finding the Helmholtz free energy:

$$\tilde{F}(T,V,N) = U_T - T\tilde{S}(U_T,V,N),$$

where $U_T = U_T(V,N)$ is the unique solution of the equation

$$T_S(U_T(V,N), V, N) = T,$$

for fixed values of N and V.

Proof: Exercise ///

Example (6.9): Let us use the second version to calculate the free energy for the monatomic ideal gas. Recall,

$$\tilde{S}(U,V,N) = Ns_0 + NR \ln\left(\left(\frac{U}{U_0}\right)^{3/2}\left(\frac{V}{V_0}\right)\left(\frac{N}{N_0}\right)^{-5/2}\right),$$

where s_0, R, U_0, V_0, N_0 are positive constants.

Then

$$\frac{\partial \tilde{S}}{\partial U} = \frac{NR}{\left(\frac{U}{U_0}\right)^{3/2}} \cdot \frac{3}{2}\left(\frac{U}{U_0}\right)^{1/2}\frac{1}{U_0}$$

$$= \frac{3NR}{2U_0} \cdot \frac{U_0}{U}$$

$$= \frac{3NR}{2U}$$

Thus

$$T_S(U,V,N) = \frac{2}{3}\frac{U}{NR}$$

and

$$U_T = \frac{3NRT}{2}.$$

Therefore,

$$\tilde{F}(T,V,N) = \frac{3NRT}{2} - TNs_0 - NRT \ln\left(\left(\frac{3NRT}{2U_0}\right)^{3/2}\left(\frac{V}{V_0}\right)\left(\frac{N}{N_0}\right)^{-5/2}\right)$$

Define

$$T_0 := \frac{2U_0}{3N_0R}.$$

Then,

$$\boxed{\tilde{F}(T,V,N) = \frac{3NRT}{2} - TNs_0 - NRT \ln\left(\left(\frac{T}{T_0}\right)^{3/2}\left(\frac{V}{V_0}\right)\left(\frac{N}{N_0}\right)^{-1}\right).}$$

What are the derivatives $\frac{\partial \tilde{F}}{\partial T}, \frac{\partial \tilde{F}}{\partial V}, \frac{\partial \tilde{F}}{\partial N}$?

Show that

$$\frac{\partial \tilde{F}}{\partial T} = -\tilde{S}(U_T(V,N),V,N),$$

$$\frac{\partial \tilde{F}}{\partial T} = -P_S(U_T(V,N),V,N),$$

$$\frac{\partial \tilde{F}}{\partial T} = \mu_S(U_T(V,N),V,N).$$